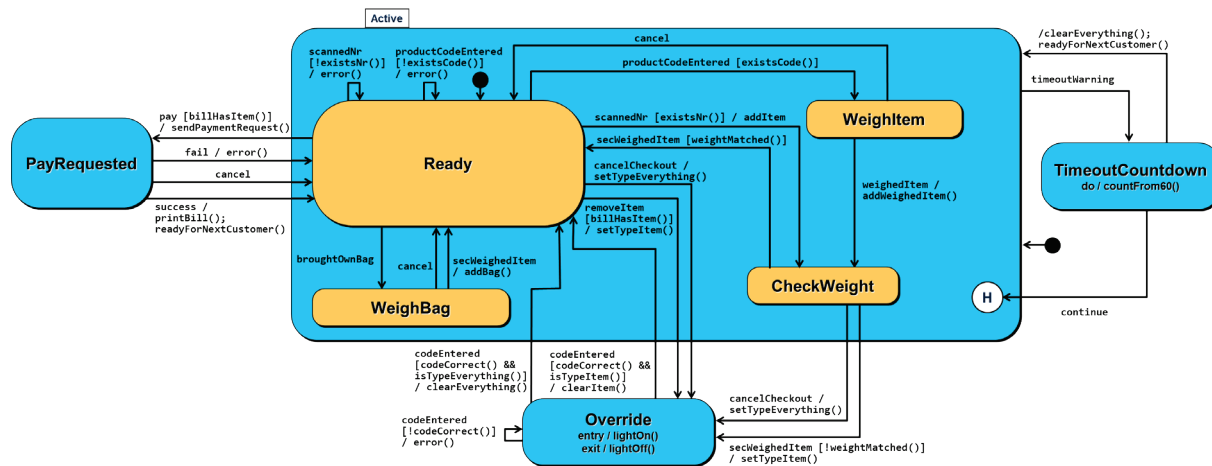


State Machine Question (100 marks)

Given the problem description below, specify the *state machine* for the software of a self-service checkout. Show the states as well as transitions including events, guards, and actions. Define composite states when needed to reduce complexity but do not use orthogonal regions. You do not need to specify parameters for events, guards, and actions. Use meaningful method names for guards and actions in your state machine. Method names must clearly show what the system needs to do or to which value a variable is set. You do not need to define the method bodies of those methods.



Alternatives: Instead of a do action in the TimeoutCountdown state, it is also possible to add the "after 60 seconds" event to the transition with the "clearEverything(); readyForNextCustomer();" actions.